

Work orders: executing the CZB roadmap with a smaller model

2026-08-27. Jonas asked whether the roadmap's execution can move to a less expensive model, and what would facilitate that. This document is the facilitation: task cards precise enough that a capable but cheaper session can execute them without design judgment, with acceptance criteria that make success checkable and stop conditions that route genuine judgment calls back to Jonas or a stronger review session. The recommendation on when to use which model is in section 1; the cards follow.

1 Recommendation on the model split

My honest view: most of the remaining work is well-specified implementation against existing patterns, and a cheaper model executes that reliably *if* the specification carries the judgment. The split I would use:

- **Cheaper model, per these cards:** A.1, A.2, A.4, B.1–B.3 constructions, C, E.1, document rebuilds, and all reruns.
- **Stronger model (or Jonas):** interpreting A.2's bias-model comparison, A.3's pass/fail verdict, B.4's transfer readout, anything where a card's stop condition fires, and a periodic review pass over completed cards (the pattern that caught the paired-difference overstatement is worth keeping: execute cheap, review strong).
- Do not hand a cheaper session an open question ("does this look right?") — hand it a card. When a card's acceptance criterion cannot be met, the correct output is a short failure report, not an improvisation.

2 Standing rules for every card (read first, they are load conditions)

- 1 **Every number that will be quoted anywhere must come from a committed script** with its output file tracked. No in-session-only analyses — this rule exists because an uncommitted analysis produced an overstated headline once already (results doc §4.3b-ii, corrected 2026-08-27).
- 2 **Run** `python tests/test_surprise.py, test_comfortzone.py, test_causation.py, test_cutin.py` **before and after each card**; all must pass. New behavior gets new property tests in the same `check()` style.
- 3 **Never modify** `src/aidriver/preferences.py` **defaults** — released behavior is frozen; new behavior goes behind flags defaulting off.
- 4 **House style** for anything written for Jonas: US English, markdown as source, Word + PDF generated (`pandoc + docs/build_pdf.py`), no period ending a heading, opinions hedged as opinions. OneDrive locks: if a build fails with `PermissionError`, write a -vN name and say so.
- 5 **Environment:** Python 3.14, torch CPU-only, no LaTeX/GPU/gh. Long jobs checkpointed and restartable; background runners never piped through `tail`.
- 6 **Provenance discipline:** cite only verified references; mark anything unverified.

3 Task cards

Card A.1 — synthetic-recovery harness

- **Goal:** prove the stage-1 estimator recovers known parameters before touching real data.
- **Build:** `replication/czb/fit_recovery.py`. Simulate the Random cut-in design (43 participants × 18 cells × 4 repetitions) from the generative model in `docs/czb_fitting_plan.md` §2 stage 1: $c_i \sim \text{LogNormal}(\mu, \sigma)$, lapse floor b (both variants), $P = b + (1-b) \cdot \Phi((x - c_i)/\sigma_{\text{resp}})$ with x = the cell's `deficit_max` from `comfortzone.czb_data.random_cutin_trials()`. Fit by MAP in torch (L-BFGS) with Laplace SEs; also fit the `a_allowed` parameterization (threshold on `a_req_max`).
- **Accept:** over 20 simulated datasets, population μ and σ recovered within 2 Laplace SEs in ≥ 18 ; per-driver c_i shrinkage visible (posterior means closer to truth than raw per-driver MLEs in RMSE). Runtime under 10 minutes total.
- **Pitfalls:** identifiability of $(c, \sigma_{\text{resp}})$ scale — fix σ_{resp} 's prior scale from the pilot ($\approx 2\,200$ deficit units); lapse and threshold trade off at C1 — check the recovery specifically at b .
- **Stop if:** recovery fails after two honest attempts — report the failure mode, do not loosen the acceptance criterion.

Card A.2 — stage-1 hierarchical fit, both bias variants

- **Goal:** the population distribution of the boundary level, and the bias-model decision.

- **Build:** replication/czb/fit_stage1.py, reusing A.1's model code on the real trials. Fit all four combinations: {deficit axis, a_req axis} × {group-level b, hierarchical b_i}. The ordered braking-expectation response enters as two nested thresholds (fitting plan §2); both levels free per roadmap §0b.
- **Report** (replication/czb/out/stage1_summary.md): population μ , σ with SEs; percentile table (50–95 in steps of 5) of the population distribution with CIs; held-out log-likelihood (leave-one-participant-out, 43 folds); posterior predictive P(intervene) per C1 cell against observed; fitted comfort and dread levels translated to a_allowed units and to steady-following THW at 20 m/s via comfortzone.field.critical_thw.
- **Accept:** all four fits converge; the summary table renders; LOPO likelihood distinguishes the bias variants or states that it cannot.
- **Stop if:** the C1 cells misfit under both bias variants (systematic sign pattern across criticalities) — that is the anticipation-leak question and needs review.

Card A.3 — the accumulator layer (review gate on its verdict)

- **Goal:** the pre-registered test: does time-integration close the within-scenario gap?
- **Build:** replication/czb/fit_stage2.py. Evidence = the per-frame deficit series (not the running max) from stimulus_field; linear accumulator with gain and noise, threshold from A.2's c_i; P(crossed by clip end) per trial via the closed-form Gaussian first-passage approximation or 200-path simulation per cell (cells share stimuli, so simulate per cell, not per trial). Motor latency fixed at 0.25 s; sensitivity refits at 0.15 and 0.35 s.
- **Accept / decision rule (pre-stated):** held-out (leave-one-criticality-out) RMSE on the 18-cell surface ≤ 0.11 counts as closing the gap (the 2D state rule's 0.120 and the design ceiling's ~ 0.10 bracket the target); > 0.13 counts as failure and is reported as evidence against the framework per the assessment.
- **Stop if:** the fitted gain wants to be so large the accumulator degenerates to the static threshold — report, since that itself answers the question.

Card A.4 — button-side validation

- **Goal:** predict the press-time densities from the Random-fitted model with one paradigm shift.
- **Build:** replication/czb/validate_button.py on czb_data.button_cutin_trials(). Fit δ once as a threshold shift and once as a gain shift; compare predicted vs observed press-time distributions per criticality (QQ plots to figures/, one page) and report which shape fits.
- **Accept:** the documented cross-paradigm excess (+0.26 at TTC6) is reproduced by the shifted model within 0.05 at the fixed-clip checkpoints.

Cards B.1–B.3 — transfer-scenario field constructions

Template for each: a loader in src/comfortzone/<scenario>.py following cutin.py's structure (role assignment appropriate to the scenario — **by instructed vehicle, not lateral span**), a field check script in replication/czb/ producing a per-criticality deficit figure, property tests, and the PS-versus-field correlation (the ps_vs_field.py pattern) as the acceptance readout.

- **B.1 cyclist overtake** (first): ego closes on a cyclist in lane; rear-end field nearly unchanged; criticality = lateral clearance 0.5/1/1.5 m, so the lateral term matters — use the actual clearance in dy, widths from the trace (cyclist ≈ 0.6 m). Accept: deficit ordered by clearance at C2–C5; cell-level PS Spearman ρ reported.
- **B.2 truck overtake / truck lateral:** ego passes a truck with clearance variants; the conflict is lateral, so P_lane runs on the *lateral closing* induced by the truck's drift (truck_lateral_movement traces) and the clearance term carries the statics. Accept: same criteria; also record whether the zero-deficit issue appears.
- **B.3 LTAP:** crossing geometry. Build arrival-time separation at the conflict zone (the intersection scenario's construction): compute each vehicle's time to the crossing-region entry/exit from the traces, and gate the collision/safety terms by predicted co-occupancy (the P_lane idea with lateral overlap replaced by zone co-occupancy). This card has real design content — **write the construction note first (1 page, the lane-entry-note pattern) and get it reviewed before coding.**
- **Stop for all three if:** a trace family's geometry does not fit the template (e.g. missing vehicle, broken dimensions beyond the documented truck case).

Card B.4 — the transfer test (review gate; do not run before A.3 and B.1–B.3)

- Fit on cut-in only (A.2/A.3 configuration frozen); predict each transfer scenario's response surface with no refit. Primary: no per-scenario shift. Secondary: one instruction shift per scenario. Comparators on identical folds:

per-scenario 2D state rules (1/TTC-analogue + the scenario's own second observable), and — once specified with Jonas — the elliptical joint on the same observables.

- Output: one table, per scenario × model: held-out corr and RMSE against (a) chance and (b) the per-scenario-refit ceiling. No verdict prose from the executing session; the verdict is a review task.

Card C — percentile sensitivity

- `replication/czb/percentile_sensitivity.py` from A.2's population distribution: implied trigger onset per stimulus for percentiles 50–95 (step 5) × `a_allowed` on the fitted comfort–dread range; report $d(\text{onset})/d(5 \text{ percentile points})$ in seconds against the percentile's own CI width. One figure, one table.

Card E.1 — closed-form versus sampled lane-entry expectation

- Monte Carlo the released binary gates under steering noise (heading random walk, $\sigma_\omega = \text{CUTIN_W_SD_MODEL}$, the model's own dt) along the 12 cut-in clips; compare the sampled conflict-probability with `lane_entry_weight`. Accept: max absolute difference reported per clip; agreement within ~0.15 through the response window validates the closed form, larger structured disagreement goes to review.

Cards E.2, E.3 and D

Deliberately not carded for cheap execution: E.2 (uncertainty term) and E.3 (policy-set field) involve model design, and D (safety audit) touches the causation runner's intervention injection. These get cards after their design notes exist.

4 Suggested session pattern

One card per session, in card order within a stage; start each session with "read `handover.md`, `docs/czb_validation_roadmap.md`, and `docs/czb_work_orders.md`, then execute card X and nothing else"; end each with the card's report, a commit whose message names the card, and one paragraph appended to the running log in `replication/czb/out/worklog.md`. Review sessions (stronger model) after A.2, A.3, and B.4 at minimum.